Raja Qamar Ilyas

Senior BIM Engineer | BIM Coordinator | BIM Specialist | ISO 19650 | PMP®

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PROFESSIONAL SUMMARY

BSI ISO 19650-certified BIM Engineer and PMP® professional with 7+ years of experience delivering highway infrastructure, digital engineering, and BIM projects across Saudi Arabia, Qatar, and Pakistan. Designed six highway and motorway projects totaling approximately 496 km and currently leading LOD 500 as-built modelling, ISO 19650 information management, and digital handover for NEOM Highway 55. Experienced in road design, utility modelling, BIM coordination, Autodesk Construction Cloud, IFC/COBie delivery, clash detection, and multidisciplinary team leadership.

7+	6	~40 km	6	LOD 500
YEARS EXPERIENCE	ROAD PROJECTS DESIGNED (~496 KM)	NEOM HWY 55 HANDOVER	INTERCHANGES DELIVERED	AS-BUILT MODELS

KEY ACHIEVEMENTS

- **NEOM Highway 55 (Packages 04 & 05):** led BIM handover of roadway assets and interchanges, including LOD 500 as-built models, COBie datasets, IFC deliverables, asset information documentation and ISO 19650-compliant digital handover.
- **Road design:** designed six highway and motorway projects in Pakistan (approx. 496 km of corridor), including Motorway M-4 (Gojra–Khanewal) and the Dera Ghazi Khan Northern By-Pass.
- **BIM coordination:** coordinated a team of approx. 10 across five disciplines (infrastructure, structure, electrical, transportation and signage)

PROFESSIONAL MEMBERSHIPS & CERTIFICATIONS

- Saudi Council of Engineers (SCE) Registered Engineer
- PMP® – Project Management Institute (PMI), USA
- CanBIM Foundation Software Skills Certification (S4)
- Certified Information Manager Level 2 Advanced – Plannerly
- BSI ISO 19650 Certified Professional
- Autodesk Certified Professional (ACP)
- Autodesk Civil 3D Building Transformation Certification

TECHNICAL SKILLS & CORE COMPETENCIES

Road Design & BIM	Autodesk Civil 3D | InfraWorks | Revit | Autodesk Vehicle Tracking
Coordination & Delivery	Navisworks (Manage & Quantification) | Revizto | Autodesk Forma | Autodesk Construction Cloud
Automation & GIS	Dynamo | Global Mapper | ArcMap / ArcGIS
Project Controls	Primavera P6 | Synchro 4D | Autodesk ReCap
Standards	ISO 19650 (BEP, MIDP, TIDP) | IFC | COBie | Saudi Road Manual | AASHTO | Dubai Road Manual
Core Competencies	Highway Geometric Design & Route Planning | Intersection & Roundabout Design | Civil 3D Corridor, Grading & Surface Modelling | Dry & Wet Utility Modelling | BIM Infrastructure Modelling & Coordination | BIM QA/QC & Model Validation | Clash Detection & Model Federation | CDE Management | 4D/5D BIM & QTO | Digital Handover & Asset Information Delivery

PROFESSIONAL EXPERIENCE

BIM Engineer (Roads) / BIM Specialist

Al-Ayuni Investment & Contracting Co. | Riyadh, KSA | March 2025 – Present

Projects: NEOM Highway 55 (Pkgs 04 & 05) • Diriyah Gate II Package 1 • SILZ

- Led BIM handover for NEOM Highway 55 Packages 04 & 05: LOD 500 as-built models, COBie datasets, IFC deliverables and asset information documentation.
- Managed BIM information in line with ISO 19650, TIDP, MIDP and CDE requirements, issuing approx. 52 models across all disciplines; administered Autodesk Construction Cloud workflows for controlled information exchange, revision management and document coordination.
- Ran federated model coordination and clash detection in Navisworks Manage across civil, structural and utility disciplines: approx. 150 clashes, of which 132 resolved and 18 approved; ran model-based QTO in Navisworks Quantification.
- Performed BIM QA/QC reviews against naming conventions, classifications and LOD requirements; prepared and validated IFC and COBie deliverables for information exchange and asset handover.
- Modelled roads in Civil 3D from LOD 300 through to LOD 500 as-built, including complex interchanges, roundabouts and junctions, plus utilities and site grading (alignments, profiles, assemblies, corridors), in accordance with the project BEP, LOD requirements and Saudi Road Manual.

- Supported BIM coordination meetings and technical reviews, tracking and resolving model issues throughout design and construction phases
- Coordinated a team of approx. 10 across infrastructure, structure, electrical, transportation and signage disciplines, keeping federated models and as-built deliverables accurate and current.
- Supported the preparation and implementation of BIM Execution Plans (BEP), information delivery plans, and project information requirements to facilitate efficient digital project delivery.
- Supported IFC validation, data verification, and digital handover processes to ensure accurate transfer of asset information to the client and operations team.

BIM Infrastructure Engineer

CDM Smith | Qatar | November 2023 – January 2025

Project: Road Western Area WA 2315 – design of road and infrastructure for the access road to Al Nakasha site from Salwa Road

- Developed and coordinated Civil 3D models for highways, access roads, roundabouts, utilities and grading to project BIM standards and LOD requirements, including alignments, profiles, assemblies, corridors, intersections and surfaces.
- Validated turning movements and geometric design compliance using Autodesk Vehicle Tracking.
- Built 3D models of dry and wet utilities (electrical, telecom, stormwater, sewer and water supply) for multidisciplinary coordination.
- Conducted clash detection, constructability reviews and federated model coordination in Navisworks Manage, resolving conflicts across civil, utility and structural disciplines.
- Generated model-based quantity take-offs, 3D solids and construction documentation: plan & profile drawings, cross-sections, grading plans and utility layouts.
- Used Autodesk Construction Cloud for BIM collaboration, document control and issue tracking; submitted BEP-compliant deliverables supporting BIM QA/QC and IFC validation.
- Produced detailed 3D and 2D deliverables including plan and profile drawings, cross-sections, grading layouts, utility layouts, and construction documentation

Geometric Designer / BIM Engineer – Infrastructure

Prime Engineering & Testing Consultants | Pakistan | September 2019 – November 2023

- Designed horizontal and vertical alignments for expressways, highways, urban arterials, collector and local roads to AASHTO and Dubai Road Manual standards (design speed, sight distance, superelevation, curve widening), across six highway and motorway projects totalling approx. 496 km.
- Designed typical cross-sections (lanes, shoulders, medians, side slopes), intersections, roundabouts and parking lots, and determined earthwork cut/fill quantities and embankment protection details.
- Prepared detailed design drawings, final tender documents and construction drawings (layout plans, plan & profile, grading plans, cross-sections), revised to client and consultant review comments.
- Provided design support during construction supervision, resolving site queries and issuing design revisions; provided in-house design support to international contractors and consultants.
- Built 3D road models in Civil 3D, InfraWorks, Revit and Navisworks; led Navisworks clash detection between roadway components and water supply, drainage and sewerage networks.
- Imported and validated survey data in Civil 3D to build accurate DTMs; extracted quantities, schedules and cut sheets from BIM models for construction and regulatory submissions; used ArcGIS for spatial analysis.
- Quality-checked designs and BIM models for accuracy, design-criteria compliance and BIM standards; coordinated planning and scheduling with architects, structural engineers and other stakeholders.

Highway & motorway projects designed (design & construction supervision)

- Lahore–Sialkot Motorway Link Highway, 4-lane (73 km)
- Motorway M-4 Gojra–Khanewal Section (126 km)
- Quetta City new roads, Packages I–IV (22 km)
- Kashmir Highway along Jhelum River (129.8 km)
- Dera Ghazi Khan Northern By-Pass (20.2 km)
- Kharan–Basima Road (125 km)

In-house support to international contractors & consultants:

- Suki Kinari Dam road relocation (4.75 km) – China Gezhouba Group
- Lowari Tunnel North & South access roads (8.2 km)
- Shounter Rattu Tunnel, AJK – DOHWA Engineering Co., Ltd.
- Peshawar–Torkham Expressway incl. Southern Link Road (84 km)

EDUCATION

Post Graduate Diploma, BIM – Construction Management | CanBIM Foundation, Canada **2025**

B.Sc. Civil Engineering Technology | Preston University, Islamabad **2019**